

APPENDIX A. TRAINING AND IMPLEMENTATION DETAILS

Training loop summary

Algorithm A1 Physics-aligned MoE training

Input: history windows X, graph sequence A, targets Y, primary regime anchors R, auxiliary wake labels W (WTB only), model parameters θ

for each minibatch do

Build node-level graph context from the precomputed graph tensors

Encode X with the directed-diffusion GRU to obtain node context $\mathbf{h}_t$

Compute gate logits Z and soft routing probabilities G

Compute expert forecasts and aggregate them with G

Evaluate L_pred

If MoE mode is active:

add L_bal on top-K routing statistics

add L_align on valid primary regime anchors

if WTB: add L_force on MPPT/pitch samples

if WTB: add L_aux on wake labels

add L_smooth on the final graph snapshot

Update θ with AdamW and mixed precision

end for

Select the checkpoint with the best validation RMSE

The wake weight is $\tilde{\mathcal{A}}_t(i, j) = \exp(-d_{ij,t}^\parallel / \alpha) \exp(-|d_{ij,t}^\perp| / \beta) \mathbb{I}_{ij,t}^{\text{cone}}$. If wind direction is missing, fall back to $\mathcal{A}^{\text{static}}(i, j) = \exp(-\|\Delta \mathbf{p}_{ij}\|_2 / d_{\max})$. Only the strongest M inbound weights are retained. The wake score and flag are $s_{ij,t}^{\text{wake}} = \sum_j \mathcal{A}_t(i, j)$ and $\mathbf{w}_{i,t} = \sum_j \mathcal{A}_t(i, j) \mathbf{w}_j$.

$$\mathbf{w}_{i,t} = \begin{cases} 1, & s_{ij,t}^{\text{wake}} \geq q_{0.75}^{\text{wake}} \wedge R_{i,t}^{\text{wtb}} \in \{1, 2\}, \\ 0, & s_{ij,t}^{\text{wake}} < q_{0.75}^{\text{wake}} \wedge R_{i,t}^{\text{wtb}} \in \{1, 2\}, \\ \emptyset, & \text{otherwise.} \end{cases}$$

For ERA5, the retained symmetric graph uses a Gaussian kernel on great-circle distances:

$$\mathcal{A}(i, j) = \exp\left(-\frac{d_{ij}^2}{2\sigma^2}\right) \mathbf{1}[j \in \mathcal{N}_{k_{\text{nn}}}(i) \text{ or } i \in \mathcal{N}_{k_{\text{nn}}}(j)],$$

where σ is the median retained neighbor distance on the training graph.

Shared constants

Model-family and supplementary evidence index

TABLE I
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Table A1. In-family model comparison used for mechanism validation.				
Model	Routing	Extra terms	Params	Role
Dense (matched)	Single dense head	None	WTB 110,012 / ERA5 103,272	Shared-mapping reference
Unconstrained MoE	Node-level soft gate	Prediction loss only	WTB 110,012 / ERA5 103,843	Routed-capacity control
Corrected routing comparator	Node-level soft gate	WTB boundary-forced: $L_{\text{bal}} + L_{\text{align}} + L_{\text{force}}$; ERA5: full corrected stack	WTB 110,012 / ERA5 103,843	Accountability comparator

Auxiliary losses

Let $\mathcal{B}$ denote the routed samples after masks, with $B = |\mathcal{B}|$. The soft importance and normalized share of expert e are $I_e = \frac{1}{B} \sum_{n=1}^B g_n^{(e)}$ and $P_e = I_e / \sum_r I_r$. The top-K load is $f_e = \frac{1}{B} \sum_{n=1}^B \mathbf{1}[e \in \text{TopK}(\mathbf{g}_n)]$, giving

$$\mathcal{L}_{\text{bal}} = E \sum_{e=1}^E f_e P_e - 1.$$

For WTB wake supervision,

$$\mathcal{L}_{\text{aux}} = \frac{1}{|\Omega_{\text{wake}}|} \sum_{(i,t) \in \Omega_{\text{wake}}} \text{BCE}(z_{i,t}^{(\text{wake})}, W_{i,t}),$$

where Ω_{wake} contains only MPPT and pitch-control samples with defined wake flags. The graph-smoothness penalty is

$$\mathcal{L}_{\text{smooth}} = \frac{\sum_{t \in \mathcal{T}_{\mathcal{B}}} \sum_{i,j} \mathcal{A}_t(i, j) \|\mathbf{g}_{i,t} - \mathbf{g}_{j,t}\|_2^2}{\sum_{t \in \mathcal{T}_{\mathcal{B}}} \sum_{i,j} \mathcal{A}_t(i, j) + \epsilon}.$$

Graph construction details

For WTB, the downstream unit vector is $\mathbf{u}_{i,t} = [\sin(\theta_{i,t} + \pi), \cos(\theta_{i,t} + \pi)]^\top$. With $\Delta \mathbf{p}_{ij} = \mathbf{p}_j - \mathbf{p}_i$, the streamwise and cross-stream distances are $d_{ij,t}^\parallel = \Delta \mathbf{p}_{ij}^\top \mathbf{u}_{i,t}$ and $d_{ij,t}^\perp = |\Delta p_{ij}^x u_{i,t}^y - \Delta p_{ij}^y u_{i,t}^x|$. A candidate edge activates when

$$\mathbb{I}_{ij,t}^{\text{cone}} = \mathbf{1}\left[d_{ij,t}^\parallel > 0 \wedge \arctan\left(\frac{d_{ij,t}^\perp}{\max(d_{ij,t}^\parallel, 10^{-6})}\right) \leq \phi\right].$$

TABLE II
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Table A3. Shared architecture, graph, and training constants used in the reported experiments.

Item	WTB	ERA5	Role
History / horizon (H, P)	(36, 24)	(36, 24)	Input and forecast lengths
Gate softmax temperature τ	0.7	0.7	Soft routing sharpness
Top-K in L_{bal}	1	1	Balancing statistic only
Hidden size d	64	64	Shared encoder width
k_{nn} candidates	8	8	Initial neighborhood size
Maximum candidate radius $d_{\max}$	1500 m	not used	WTB proximity fallback / candidate filter
Wake-cone half-angle ϕ	25°	not used	WTB downstream activation
Streamwise decay α	1200 m	not used	WTB wake attenuation
Cross-stream decay β	400 m	not used	WTB wake attenuation
Retained inbound edges M	5	symmetric k_{nn} graph	Final graph sparsity
Optimizer	AdamW	AdamW	Shared optimizer
Learning rate	2×10^{-3}	2×10^{-3}	Shared schedule
Weight decay	10^{-4}	10^{-4}	Shared regularization
Batch size	16	16	Shared mini-batch size
Gradient clipping	1.0	1.0	Shared training stabilization

Dataset-specific thresholds and routing weights

TABLE III
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Table A4. Dataset-specific regime thresholds used to construct routing anchors.

Dataset	Threshold	Value and meaning
WTB	u_{idle}	3.0 m s ⁻¹ , idle / active split
WTB	u_{rated}	10.5 m s ⁻¹ , MPPT / pitch boundary
WTB	p_{th}	2.0 deg, pitch-angle threshold
WTB	$q_{0.75}^{\text{wake}}$	1.0690, upper-quartile wake-score threshold
ERA5	$\varepsilon_{\text{sshf}}$	169,116, 10th percentile of sshf on the training split
ERA5	$q_{0.95}^{\Delta}$	458,287.125, 95th percentile of $ \Delta \text{sshf} $ on the training split

TABLE IV
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Table A5. Active routing-loss weights in the reported corrected models.

Weight	WTB	ERA5	Role
λ_{bal}	1000	0.01	Expert-usage stabilization
λ_{align}	5000	0.2	Coarse regime alignment
λ_{force}	10000	0	Boundary-focused forcing
λ_{aux}	250	0	Wake-sensitive auxiliary routing
λ_{smooth}	0.05	0.05	Local routing coherence

Statistical claim boundaries

Table A6 records the reviewer-facing statistical boundary used for wording. The RMSE price versus Graph WaveNet is FDR-significant, gate-alignment deltas versus the full physics-aligned MoE are positive but not FDR-significant, and boundary-window quantile reserve comparisons have bootstrap intervals crossing zero. These tests are why the main text uses price, diagnostic, and bounded-claim language rather than forecast- or reserve-superiority wording.

TABLE V
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Table A6. Statistical claim boundaries used for reviewer-facing wording.

Claim	Estimate	95% CI	p_{BH}	Wording consequence
Accuracy price vs Graph WaveNet	9.029	—	<0.001	Boundary router is significantly worse on headline RMSE
Boundary-band accuracy price	17.758	—	0.001	The price also appears in the boundary window
Gate NMI vs full physics-aligned MoE	0.040	[0.008, 0.071]	0.066	Positive but not FDR-significant; cite as bounded mechanism contrast
Gate ARI vs full physics-aligned MoE	0.035	[-0.000, 0.078]	0.126	Positive but not FDR-significant; avoid superiority wording
Boundary quantile cost vs GWN physical bin	-0.263M	[-10.704M, 9.782M]	—	CI crosses zero; reserve cost should remain a diagnostic claim
Boundary quantile violation vs GWN physical bin	0.003	[-0.011, 0.020]	—	CI crosses zero; no universal reserve-policy optimality claim

Outcome-channel sanity audit

Table A7 adds a bounded check for the shared-anchor concern. It does not use pitch-threshold labels to score the contrast: validation data define a wind-speed-bin power curve, and the test-set comparison is restricted to 9.5–11.5 m s⁻¹ boundary anchors with fine wind-bin adjustment. The result asks whether the recovered gate separates samples with different future active-power response, not whether it discovers a regime without anchors.

TABLE VI
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Table A7. Outcome-channel sanity audit for the shared-anchor concern.

Check	Five-seed summary	Interpretation
Boundary cells per seed	3276 MPPT / 10465 pitch	9.5–11.5 m s ⁻¹ test anchors
Future mean power	100.0 ± 17.5 kW	Pitch-gate minus MPPT-gate after wind-bin adjustment
Power-curve residual	100.0 ± 17.5 kW	Validation wind-bin power curve only; no pitch label used
Future power ramp	81.8 ± 24.7 kW	Mean late-horizon minus early-horizon power
Anchor-time Patv	-44.1 ± 32.5 kW	Current active power is not driving the same positive contrast
Residual claim boundary	sanity check only	Mitigates circularity concern; does not prove anchor-free discovery

External-site deployment-gate audit

Table A8 records the go/no-go interpretation used for the Kelmarsh/Penmanshiel evidence. Each row maps a failed or incomplete external signal to the deployment action that follows. The table therefore supports a bounded claim: the external experiments are a screening protocol for new wind farms, not evidence that the WTB gate can be transferred directly.

TABLE VII
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Table A8. External-site deployment-gate audit and wording boundary.			
Gate	Observed external evidence	Go/no-go rule	Claim consequence
Cross-site routing criterion	80/80 runs complete; mean NMI 0.4877 below threshold 0.50; mean ARI 0.5112	Held-out NMI $\geq$ 0.50 and balanced accuracy $\geq$ 0.50 after parameters are frozen	No-go for cross-farm router interpretation; cite as negative boundary-condition evidence
Local boundary recalibration	Default test NMI 0.1324 $\rightarrow$ recalibrated 0.1491 (delta +0.0167)	Rated wind, pitch threshold, boundary band, and gate-map selected on calibration only	Local threshold transfer is insufficient; re-estimate before use
Small-window adaptation	40/40 routing runs adapted; chronological balanced accuracy 0.4787 below 0.50	Small calibration windows must still pass the frozen held-out routing criterion	Calibration alone does not authorize external reserve use
Sensor and boundary support	Penmanshiel-to-Kelmarsh leave-one pitch-feature coverage 0.0000 and effective boundary cells 0	Pre-declared calibration window with enough boundary cells, active power, availability mask, and pitch/proxy overlap	No physical-router interpretation without observability
External reserve-use decision	Upstream gates do not pass before reserve allocation is evaluated	Transition-window shortage and violation improve at acceptable reserve-energy cost	Withhold gate-bin reserve use outside WTB; report a deployment protocol only

Early-warning detection consequence

Table A9 reports the cell-count version of the label-degradation audit. Counts are turbine-time cells per seed inside the six-step MPPT-to-pitch window; they are not MWh, currency, or dispatch-cost estimates. The purpose is narrower: it shows how many early pitch-window cells the gate preserves when a threshold-label rule is delayed, incomplete, or noisy.

TABLE VIII
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Table A9. Early-warning detection consequence under degraded threshold labels.			
Condition	Gate detected / missed	Degraded rule detected / missed	Recovered cells
Delay, 1 step	710.6 $\pm$ 20.6 / 29.4 $\pm$ 20.6	485.0 $\pm$ 0.0 / 255.0 $\pm$ 0.0	225.6 $\pm$ 20.6
Delay, 3 steps	710.6 $\pm$ 20.6 / 29.4 $\pm$ 20.6	282.0 $\pm$ 0.0 / 458.0 $\pm$ 0.0	428.6 $\pm$ 20.6
Delay, 6 steps	710.6 $\pm$ 20.6 / 29.4 $\pm$ 20.6	145.0 $\pm$ 0.0 / 595.0 $\pm$ 0.0	565.6 $\pm$ 20.6
50% label availability	710.6 $\pm$ 20.6 / 29.4 $\pm$ 20.6	376.0 $\pm$ 10.4 / 364.0 $\pm$ 10.4	334.6 $\pm$ 28.1
25% label availability	710.6 $\pm$ 20.6 / 29.4 $\pm$ 20.6	179.0 $\pm$ 7.9 / 561.0 $\pm$ 7.9	531.6 $\pm$ 22.9
Sensor noise, strongest	710.6 $\pm$ 20.6 / 29.4 $\pm$ 20.6	482.8 $\pm$ 11.0 / 257.2 $\pm$ 11.0	227.8 $\pm$ 20.9

Reserve-policy claim-boundary audit

Table A10 consolidates the reserve evidence used for wording. It separates the same-model boundary-window diagnostic from claims that the experiments do not support. Costs are

normalized reserve-energy proxy units, not currency, market prices, or security-constrained dispatch costs.

TABLE IX
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Table A10. Reserve-policy claim-boundary audit.		
Boundary	Evidence	Wording rule
Same-model boundary reserve effect	At $\rho=10$, gate-bin vs same-router global: Delta cost -3.55M, Delta viol. -0.0138, reserve +1.85M, shortage -0.54M.	Claim a same-predictor transition-window diagnostic; do not present this as a cross-backbone reserve win.
Physical-bin quantile comparator	Boundary/physical-bin 83.78M, viol. 0.0880; GWN/physical-bin 84.31M, viol. 0.0931; Boundary/gate-bin 84.58M, viol. 0.0900.	Say physical-bin baselines are competitive and sometimes lower-cost; avoid gate-bin optimality wording.
Cost-ratio applicability	$\rho=2$ inactive; $\rho=5$ to 10 lowers boundary cost and violation; $\rho=20$ narrows; $\rho=50$ favors global (+5.93M, viol. +0.0035).	Claim moderate-cost transition-window value only; do not assert a universal shortage-penalty policy.
Full-sample system value	Full sample at $\rho=10$: GWN/global 464.07M, viol. 0.0901; Boundary/gate-bin 481.36M, viol. 0.1214.	Do not claim system-wide dispatch value or reserve superiority; keep the consequence bounded to the boundary slice.
Seed-level uncertainty	Full-sample paired total-cost delta +17.28M, 95% CI [-52.09M, +87.95M], perm. $p=0.752$.	Use bounded diagnostic language; do not cite the reserve audit as a statistically settled improvement.
Operational scope	Costs are normalized reserve-energy proxy units from validation-frozen shortfall quantiles; OPF, unit commitment, delivery constraints, market clearing, and prices are excluded.	Use as a screening audit for reserve exposure, not as a market or security-constrained dispatch study.

Engineering-unit reserve-value translation

Table A11 provides the engineering-unit translation of the main reserve audit. The conversion uses the WTB active-power unit (kW) and the cache time step ($\Delta t = 1/6$ h), so reserve and shortage totals become rolling forecast-cell MWh-equivalent values. The EUR column is a scenario translation under an assumed reserve carrying cost of 100 EUR/MWh. It is included to make the operational scale legible, not to claim market settlement, OPF, unit commitment, or security-constrained dispatch value.

TABLE X
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Table A11. Engineering-unit reserve-value translation at $\rho = 10$.

Comparison	Δ reserve MWh-eq.	Avoided shortage MWh-eq.	Δ cost MWh-eq.	Δ EUR at 100/MWh	Wording
Boundary gate-bin vs same-router global	+1846.9	+539.8	-3551.4	-355k	Use as bounded boundary- window value, not cross- backbone superiori- ty.
Boundary gate-bin vs GWN physical-bin	+2319.2	+205.6	+262.9	+26k	Shows gate-bin is close to a strong physical- bin compara- tor; not a lower-cost claim.
Boundary gate-bin vs GWN global full sample	+3148.2	-1413.5	+17283.2	+1728k	Blocks system- wide dispatch or full- sample reserve- superiority wording.

MWh-eq. denotes forecast-cell MWh-equivalent accounting from kW active-power shortfall and $\Delta t = 1/6$ h. EUR values are scenario translations under an assumed reserve carrying cost of 100 EUR/MWh; they are not market-settlement, OPF, or unit-commitment results.